

WILHELMINADORP, Netherlands

WMO: 063230

Lat: 51.5260N

Lon: 3.8836E

Elev: 1

StdP: 101.31

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.8	-3.1	-8.7	1.8	-3.4	-6.3	2.2	-2.4	15.8	9.5	14.3	9.1	4.6	80	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.8	27.9	19.8	25.5	18.8	23.6	18.0	20.8	26.0	19.7	24.1	18.8	22.5	3.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.7	23.1	18.1	13.0	21.8	17.3	12.4	20.8	60.0	25.9	56.4	24.2	53.3	22.5	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.5	10.8	9.4	-7.6	32.2	2.9	2.7	-9.7	34.1	-11.4	35.7	-13.0	37.2	-15.2	39.2
			-8.0	23.1	2.8	1.0	-10.0	23.9	-11.7	24.5	-13.2	25.1	-15.3	25.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	3.9	4.5	6.8	10.3	13.5	16.0	17.9	18.0	15.6	12.0	7.9	4.7
	DBStd	5.98	3.82	3.59	3.29	3.48	3.16	2.83	2.78	2.42	2.43	3.05	2.98	3.68
	HDD10.0	750	190	155	107	38	5	0	0	0	0	16	74	166
	HDD18.3	2764	447	387	358	241	153	81	42	36	87	198	313	422
	CDD10.0	1102	1	1	7	47	114	180	244	247	169	77	12	2
	CDD18.3	76	0	0	0	1	4	11	28	25	6	1	0	0
	CDH23.3	489	0	0	0	6	34	83	191	145	28	2	0	0
	CDH26.7	119	0	0	0	0	4	22	52	37	4	0	0	0
Wind	WSAvg	4.7	5.6	5.4	5.0	4.6	4.5	4.1	4.0	3.9	4.1	4.7	4.8	5.4
Precipitation	PrecAvg	858	70	63	46	41	57	72	81	83	82	82	93	86
	PrecMax	1117	170	125	119	81	112	134	172	166	209	180	147	172
	PrecMin	608	5	19	13	0	17	22	14	14	17	15	9	19
	PrecStd	118	37	28	26	22	27	33	43	46	52	37	34	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	13.9	17.4	24.2	26.8	29.7	30.7	30.3	26.2	22.4	16.1	13.3
		MCWB	11.2	10.2	11.8	14.8	18.4	20.7	20.6	20.6	19.5	17.2	13.8	12.0
	2%	DB	11.4	12.0	14.9	21.1	23.6	25.6	27.9	27.0	23.5	18.8	14.4	12.2
		MCWB	10.3	9.9	11.1	14.1	16.5	19.1	19.7	20.1	18.2	15.7	12.8	11.0
	5%	DB	10.1	10.7	12.8	18.5	21.4	23.0	25.3	24.5	21.4	17.3	13.2	11.0
		MCWB	9.0	9.1	10.3	12.9	15.6	17.8	19.1	18.8	17.2	15.1	11.8	9.9
	10%	DB	9.0	9.4	11.5	15.9	19.1	21.1	23.2	22.5	19.6	16.2	12.0	9.8
		MCWB	8.0	8.2	9.4	11.7	14.7	17.1	18.3	17.8	16.1	14.3	10.7	8.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	11.3	13.0	16.3	20.1	21.9	22.1	22.2	20.2	17.6	14.1	12.4
		MCDB	12.4	12.8	15.8	21.9	25.1	27.4	28.5	27.9	24.8	21.0	15.4	13.0
	2%	WB	10.4	10.2	11.7	14.6	17.7	19.7	20.7	20.6	18.8	16.2	13.0	11.2
		MCDB	11.3	11.5	13.8	19.7	21.3	24.3	25.9	25.5	22.2	18.2	14.2	12.0
	5%	WB	9.1	9.2	10.6	13.3	16.2	18.5	19.6	19.5	17.7	15.4	11.9	10.0
		MCDB	9.9	10.4	12.4	17.8	20.4	22.1	24.1	23.3	20.4	17.0	13.1	10.9
	10%	WB	8.1	8.2	9.6	12.1	15.0	17.4	18.7	18.4	16.9	14.5	10.8	8.8
		MCDB	9.0	9.3	11.3	15.4	18.6	20.5	22.4	21.5	19.3	16.0	11.8	9.6
Mean Daily Temperature Range	5% DB	MDBR	3.9	4.8	6.2	7.9	7.7	7.9	7.8	7.8	6.9	5.8	4.5	3.9
		MCDBR	4.5	6.5	8.6	12.0	11.6	11.4	11.7	11.2	9.7	7.0	5.2	4.5
		MCWBR	4.3	5.0	5.6	6.1	6.0	6.4	5.5	5.3	5.1	4.6	4.4	4.3
	5% WB	MCDBR	4.5	5.6	7.6	10.8	10.9	10.2	10.3	9.8	8.4	6.6	5.2	4.6
		MCWBR	4.4	4.6	5.4	5.8	6.3	6.2	5.5	5.3	5.0	4.6	4.6	4.5
Clear-Sky Solar Irradiance	taub	0.336	0.356	0.391	0.418	0.426	0.422	0.427	0.418	0.395	0.372	0.349	0.332	
	taud	2.360	2.343	2.252	2.212	2.238	2.269	2.263	2.306	2.359	2.402	2.375	2.339	
	Ebn at Noon	664	756	795	818	829	835	823	810	785	730	641	600	
	Edh at Noon	64	84	110	128	131	128	127	116	98	79	63	56	
All-Sky Solar Radiation	RadAvg	0.74	1.44	2.66	4.34	5.14	5.55	5.22	4.46	3.27	1.87	0.87	0.56	
	RadStd	0.07	0.27	0.34	0.40	0.38	0.42	0.56	0.37	0.24	0.16	0.09	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (84 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page